

Carlos Manuel Martínez Vázquez

Computer Science Student @ Monterrey Institute of Technology

✉ c.mtnzvzqz@gmail.com

☎ +52 55-7465-6116

🌐 c-vzqz

🔗 CsVazquezz

🌐 csvazquez.com

SUMMARY

Computer Science student and Quantitative Researcher specializing in the mathematical foundations of AI, geometric data analysis, and structural modeling. Experienced in bridging stochastic machine learning architectures, heuristic multi-agent systems, and convex optimization frameworks to map non-linear properties in high-dimensional spaces. Aspiring to pursue an MSc/PhD in Applied Mathematics to research abstract system invariances, statistical convergence, and functional optimization to advance deep learning theory for high-dimensional data representation.

EDUCATION

Monterrey Institute of Technology

B.S. in Computer Science and Technology (GPA: 3.7/4.0)

Honors: Academic Excellence Scholarship.

Querétaro, México

Aug 2023 – June 2027

EXPERIENCE

Quantitative Researcher Intern

Moreton Capital Partners

Mar 2026 – Present

Mexico City, México

Architecting multi-agent heuristic systems operating as non-linear layers atop production-level deterministic models to optimize systematic commodity strategies and signal blending.

Formulating mathematically rigorous meta-models to map high-dimensional financial datasets, enforcing statistical constraints and regularizers to ensure structural stability and mitigate alpha decay.

Bridging stochastic machine learning outputs with classical convex optimization frameworks, integrating agentic signals into multi-horizon risk-parity and regime-aware allocation models.

Lead Researcher | *Model-Agnostic Infrastructure for Unstructured Financial Text*

Dec 2025 – Present

Working Paper

Formalizing an end-to-end, model-agnostic architectural blueprint for processing high-dimensional, unstructured financial text, ensuring structural invariance against shifting baseline language architectures.

Designing a modular data engineering pipeline that decouples stochastic noise-filtering mechanics from downstream deep learning systems, optimizing input embedding spaces to ensure future-proof compatibility.

Formulating a novel, domain-specific reward function and objective metric that maps linguistic signal extraction directly to portfolio optimization, independent of the underlying transformer or LLM paradigm.

RESEARCH & PROJECTS

Hierarchical Commodity Portfolio Construction

Mar 2026

Developed a two-stage backtesting engine for 10 commodity sectors over 162 weekly periods with SLSQP min variance optimization algorithm that achieved max 7.7% annual volatility.

Enforced KKT optimality to maintain stable weights and model transactions and achieved a 4.02 Adjusted Sharpe (Andrew W Lo 2002) and 2.1 Calmar Ratio by implementing Ledoit-Wolf shrinkage to regularize covariance matrices.

OrbitGrow: Martian Plant Health Monitoring | *Syngenta @ START Hack*

Mar 2026

Engineered a real-time computer vision system to monitor crop health in Martian greenhouses, utilizing AWS SageMaker for autonomous stress detection and awarded Top 5 Finalist out of 30 teams in the Syngenta Challenge at START Hack St. Gallen, Switzerland.

Developed a multi-modal data pipeline to process environmental telemetry and spectral imagery, enabling automated nutrient deficiency diagnosis in extraterrestrial environments.

AquaHub: Predictive Drought Modeling & Crowdsourcing

Feb 2026

Engineered a predictive ML model for drought detection in CDMX, processing 960+ training samples to identify water scarcity hotspots and validated a 5-day sliding window architecture, achieving 70–85% accuracy in preliminary drought event detection. Designed a validation pipeline to cross-reference crowdsourced reports against meteorological norms, covering a demographic of 17.7M residents.

Bloomly: Global Bloom Detection System | *@ NASA Space Apps Hack*

Oct 2025

Engineered a LightGBM-based predictive model leveraging multi-spectral satellite and meteorological data and conducted feature selection and dimensionality reduction across 44 ecological indicators to improve classification precision and evaluated model performance via AUC and F1 metrics under temporal cross-validation for predictive stability.

TECHNICAL SKILLS

Languages & Quant: Python, C++, R, SQL, JavaScript, scikit-learn, LightGBM, pandas, NumPy, PyTorch.

Dev & Cloud: Snowflake, React, Next.js, Node.js, Flask, PostgreSQL, AWS, Git. **Spoken:** Spanish (Native), English (C1).